LogicScope S3 / 8-channel USB digital logic analyzer.
Inputs 0-5V digital only. Common ground required.
1 MS/s finite-capture design target; measurement pending.
55x40mm, two copper layers. See engineering report.

GND

VUSB

+3V3

01 Controller and Acquisition 1

File: 01_Controller_and_Acquisition_1.kicad_sch

02 Digital Input Protection 1

File: 02_Digital_Input_Protection_1.kicad_sch

03 Digital Input Protection 2

File: 03_Digital_Input_Protection_2.kicad_sch

04 USB and Power 1

File: 04_USB_and_Power_1.kicad_sch

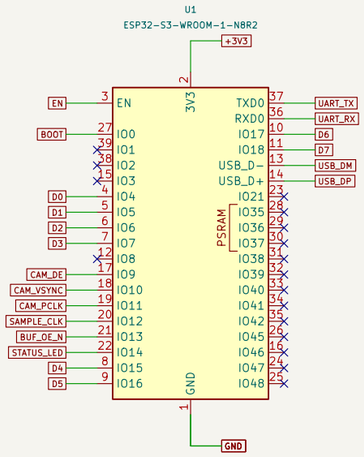
05 Test Points 1

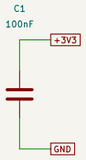
File: 05_Test_Points_1.kicad_sch

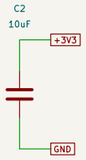
Sheet: /
File: LogicScope_S3.kicad_sch

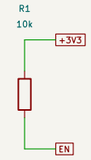
Title: LogicScope S3 | LS-S3-A0

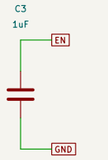
Size: A3	Date: 2026-11-09	**Rev: A.0 ENGINEERING PROTOT**
KiCad E.D.A. 10.0.6		Id: 1/6

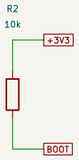


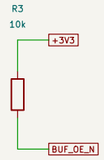


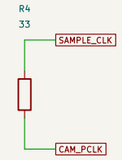


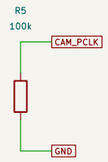


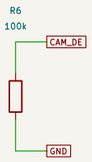


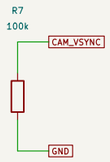




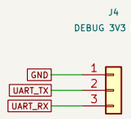










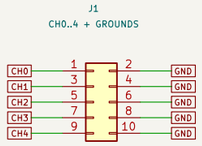


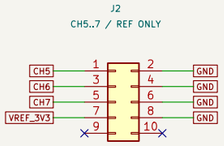
Eight buffered signals enter CAM_DATA[7:0] concurrently via GPIO matrix.
GPIO4/5/6/7/15/16/17/18 = D0..D7. GPIO12 -> 33R -> GPIO11 = sampling clock.
GPIO9/10 drive and internally loop back DE/VSYNC. GPIO13 enables buffer (active LOW).
Native USB GPIO19/20. Unused module pins are intentionally unconnected; N8R2 Quad PSRAM is not required for initial SRAM capture.

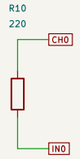
Sheet: /01 Controller and Acquisition 1/
File: 01_Controller_and_Acquisition_1.kicad_sch

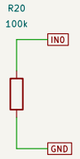
Title: LogicScope S3 | Controller and Acquisition

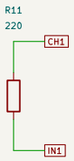
Size: A3	Date: 2026-11-09	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 2/6

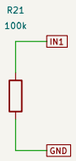


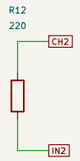


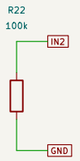


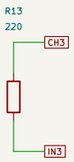


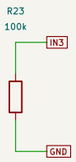


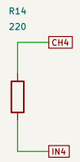


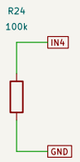


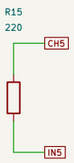


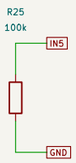


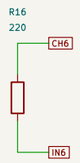


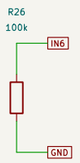


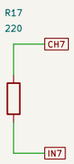


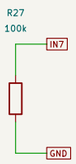


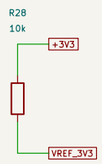


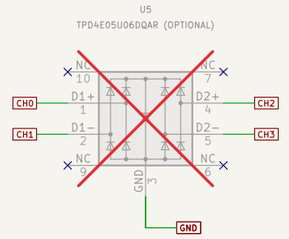






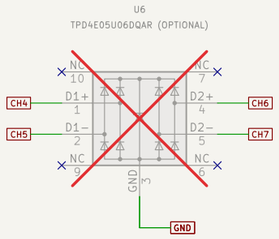


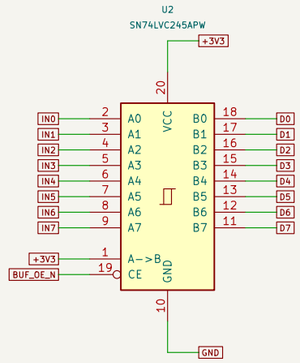


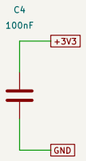


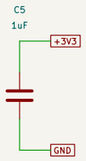
External inputs: 0..5 V DIGITAL ONLY; 220R series + 100k pulldown each.
U2 at 3.3V, DIR HIGH fixes A inputs -> B outputs; OE pullup disables outputs during reset.
U5/U6 optional DNP low-capacitance ESD footprints; NC pins intentionally unused.
Header J2 pin 7 is a 10k-isolated reference, not a target power source. J2 pins 9/10 intentionally NC.

Sheet: /02 Digital Input Protection 1/		
File: 02_Digital_Input_Protection_1.kicad_sch		
Title: LogicScope S3 | Digital Input Protection		
Size: A3	Date: 2026-11-09	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 3/6







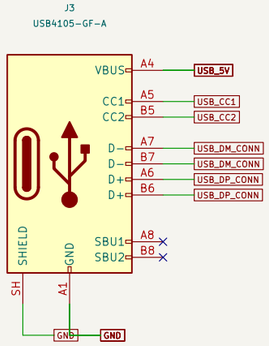


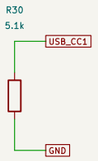
External inputs: 0..5 V DIGITAL ONLY: 220R series + 100k pulldown each.
U2 at 3.3V, DIR HIGH fixes A inputs -> B outputs; OE pullup disables outputs during reset.
U5/U6 optional DNP low-capacitance ESD footprints; NC pins intentionally unused.
Header J2 pin 7 is a 10k-isolated reference, not a target power source. J2 pins 9/10 intentionally NC.

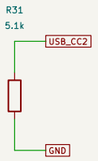
Sheet: /03 Digital Input Protection 2/
File: 03_Digital_Input_Protection_2.kicad_sch

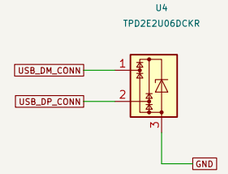
Title: LogicScope S3 | Digital Input Protection

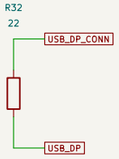
Size: A3	Date: 2026-11-09	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 4/6

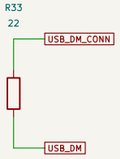


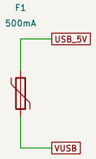


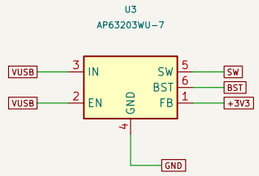










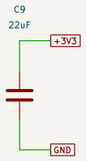


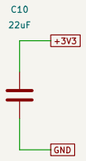


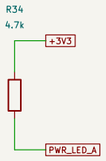


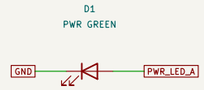


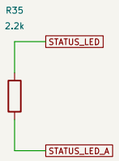


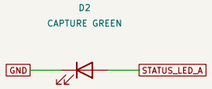












USB 2.0 FULL SPEED only. D+ = GPIO20, D− = GPIO19. Both plug orientations wired.
USB-C CC1/CC2 each have 5.1k Rd. No target-power output. Buck input is USB only.
AP63203 fixed 3.3V: FB senses output, VIN/EN on fused 5V, 100nF BST-to-SW.
SBU pins unused. USB shield bonded to GND; USB_5V test point is not an accessory output.

Sheet: /04 USB and Power 1/
File: 04_USB_and_Power_1.kicad_sch

Title: LogicScope S3 | USB and Power

Size: A3	Date: 2026-11-09	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 5/6

TP1
USB_5V



TP2
VUSB



TP3
+3V3



TP4
GND



TP5
EN



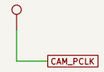
TP6
USB_DP



TP7
USB_DM



TP8
CAM_PCLK



TP9
CAM_DE



TP10
CAM_VSYNC



TP11
BUF_OE_N



All signal names are global nets. Logic is 3.3 V unless explicitly marked. See engineering documentation.

Sheet: /05 Test Points 1/
File: 05_Test_Points_1.kicad_sch

Title: LogicScope S3 | Test Points

Size: A3	Date: 2026-11-09	**Rev: A.0 PROTOTYPE**
KiCad E.D.A. 10.0.6		Id: 6/6